



AI-Based Video Analytics

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CTDOT/UCONN

A partnership in Safety

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AI-based video analytics for
intersection safety and operations

USDOT SS4A Grant - connected
video analytics system with 5G for
safe and smart campus



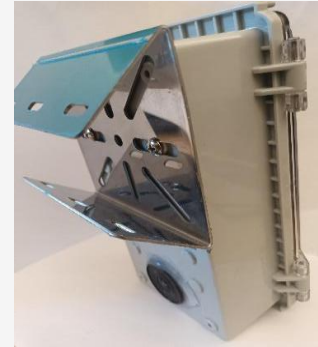
AI-Based Video Analytics at Intersections

CCTRP Project 23-01

Expanding Capabilities of AI Based Video Analytics for Intersection Safety and Operation for CTDOT



Storrs Road & N. Eagleville Road, Storrs, CT



Custom
Equipment at
Intersections

Total Cost
<\$100 each



Storrs Road & Mansfield Road, Storrs, CT



N. Eagleville Road & Discovery Dr. Storrs, CT

AI-Based Video Analytics at Intersections

Operation

- Object detection and trajectory tracking
- Vehicle travel speed
- Traffic volumes by mode (vehicles vs. pedestrians)
- Traffic volumes by travel direction (e.g. thru traffic volumes, left/right-turn volumes)

Safety

- Conflicts by approach, time period, and maneuver
- Conflicts by type (vehicle-vehicle, vehicle-ped)
- Calculation of surrogate measure/near miss (Time-To-Collision (TTC) & Post Encroachment Time (PET))



Time To Collision (TTC)

Represents the time remaining to prevent a collision

- Applying brakes
- Steering away
- Other preventive action

Measured precisely as the difference between the time the road user takes action to the point where the collision can occur

Safety Considerations

Short TTC → less time to react; higher collision risk.

Long TTC → more time to brake or maneuver.



Post Encroachment Time (PET)

The difference between the time when the first road user leaves a point or conflict area and the second road user reaches that same point or conflict area

Near-zero PET → near-miss or potential collision

PET = 0 → actual collision



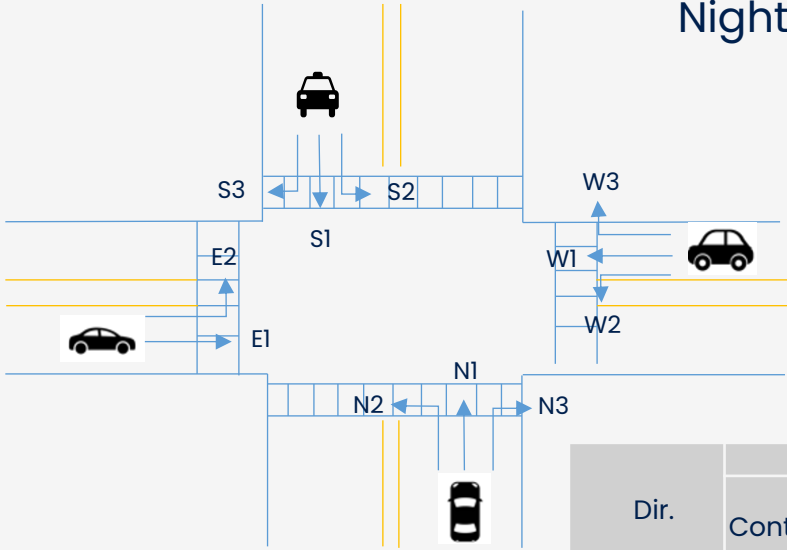
Operation – Object Detection and Traveling Speed



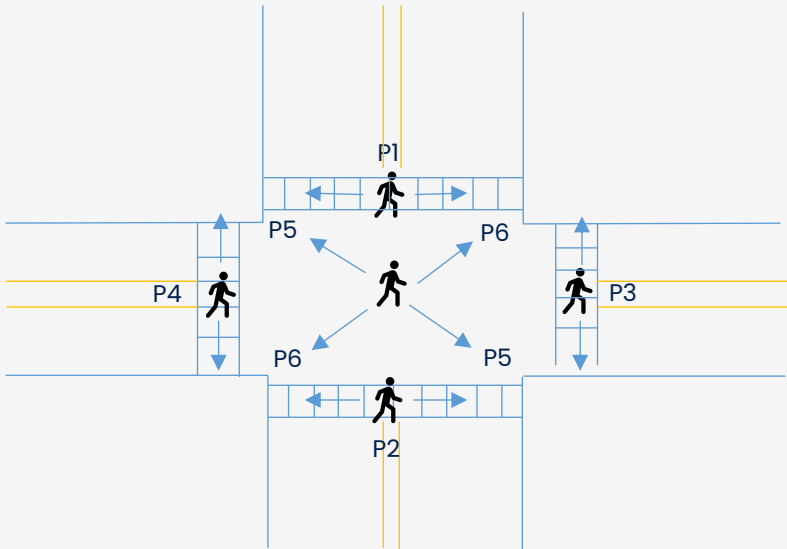


Operation – Object Detection and Traveling Speed

Day: 6:00 AM – 12:20 PM
Night: 12:34 AM – 6:00 AM



Dir.	Day			Night		
	Cont.	Speed [Avg; Min; Max]		Cont.	Speed [Avg; Min; Max]	
S1	457	19; 14; 31	104	22; 15; 32		
S2	465	10; 8; 14	142	10; 8; 14		
S3	423	10; 8; 14	127	10; 8; 12		
N1	502	19; 14; 29	194	19; 14; 31		
N2	255	15; 12; 20	106	16; 13; 19		
N3	351	9; 7; 13	194	9; 7; 12		
E1	1,344	14; 11; 20	725	14; 11; 20		
E2	565	12; 11; 16	172	13; 11; 16		
W1	931	19; 15; 25	121	17; 15; 22		
W2	124	15; 12; 20	23	13; 11; 16		
W3	335	12; 10; 17	40	12; 10; 14		



Dir.	Count	
	Day	Night
P1	26	16
P2	5	0
P3	109	26
P4	172	47
P5	0	0
P6	2	0



Safety – Time to Clear

Frame: 2176

TTC_min: 11.74 Seconds



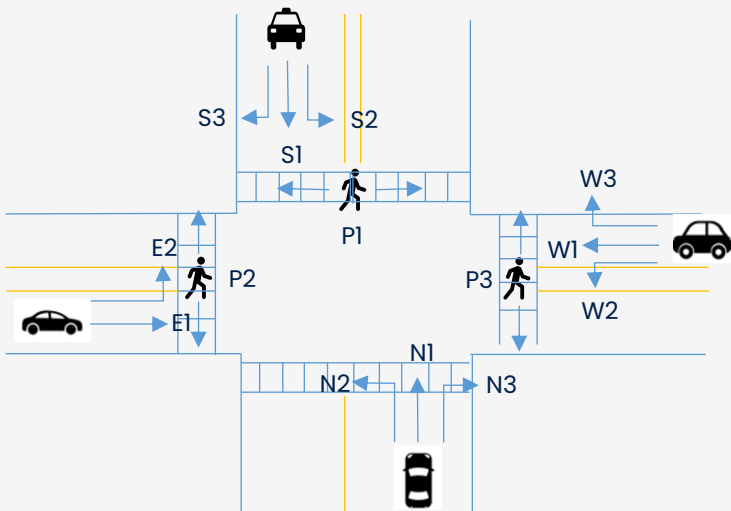


Safety – Post Encroachment Time

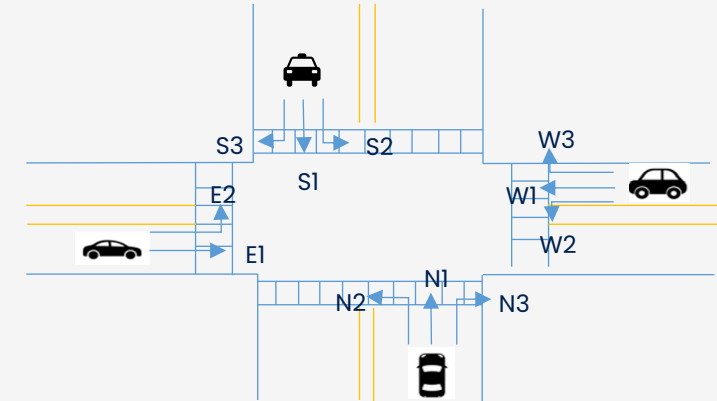




Collected Safety Metrics



Conflict Type	TTC		PET	
	$\leq 2s$	$(2s - 5s]$	$\leq 3s$	$(3s - 6s]$
S3 vs. P1	13	7	20	7
W3 vs. P1	13	22	8	16
E2 vs. P1	0	0	1	4
N2 vs. P2	0	0	0	2
S3 vs. P2	0	1	1	0
N3 vs. P3	0	0	0	0
S2 vs. P3	0	0	0	0
W3 vs. P3	0	0	0	0



Conflict Type	TTC		PET	
	$\leq 2s$	$(2s - 5s]$	$\leq 3s$	$(3s - 6s]$
S1 vs. S1	11	41	79	40
S1 vs. N2	10	11	23	17
S2 vs. N3	17	28	47	47
S2 vs. N1	28	50	40	67
N2 vs. S3	1	0	19	13
W2 vs. S3	14	13	11	19
N1 vs. W3	13	13	11	4
N3 vs. E1	10	41	34	55
E1 vs. W2	44	34	34	43
E2 vs. W3	10	14	19	29
E2 vs. W1	52	88	43	55



SS4A – Live 5G Connected Video Intersection Monitoring

- **Develop and deploy AI-based algorithms for real-time analysis of vehicle and pedestrian safety at intersections across the UConn Storrs campus.**
- **Integrate 5G and re-identification (Re-ID) technologies to create a smart, interconnected traffic monitoring network.**
- **Track common vehicles and identify risky behaviors across multiple intersections.**
- **Build a real-time dashboard to visualize:**
 - **Road user volumes and trajectories**
 - **Vehicle speeds and traffic capacity**
 - **Conflict indicators**
 - **Near-miss heat maps**
 - **Safety performance metrics**
 - **Additional mobility and safety analytics**



SS4A Grant – Live 5G Connected Video Intersection Monitoring

UConn Police Department Provides Access to Video Data at Four Intersections

- North Eagleville Road & Discovery Drive
- North Eagleville Road & CT-195
- CT-195 & Mansfield Road
- CT-195 & Bolton Road





Multi-Camera Vehicle Tracking

Unique IDs
Across
Tracking
Locations

Multi-Intersection Vehicle Tracking

YOLOv8 + ByteTrack

Detection → Tracking → Cross-Camera ReID → Analytics



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Thank You!